

NPTEL Course on "Computer System Optimization for ML"

Lecture #48

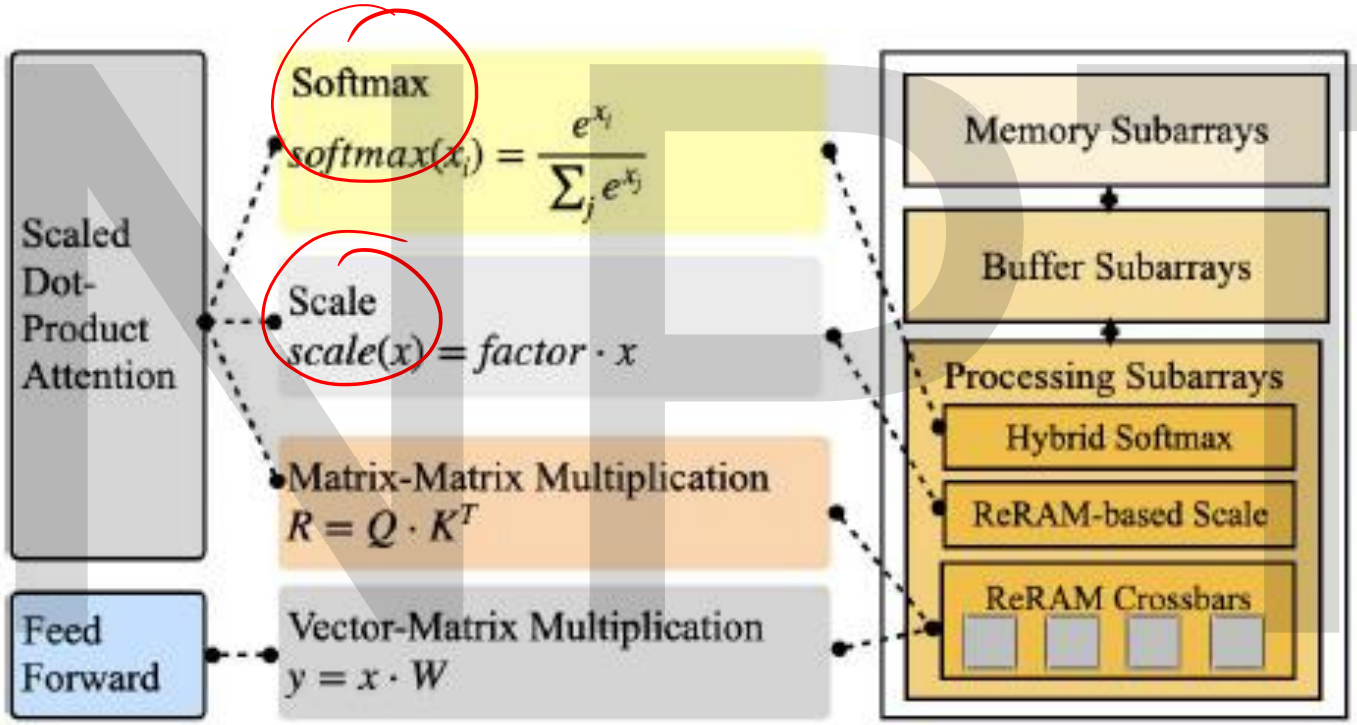
Agenda

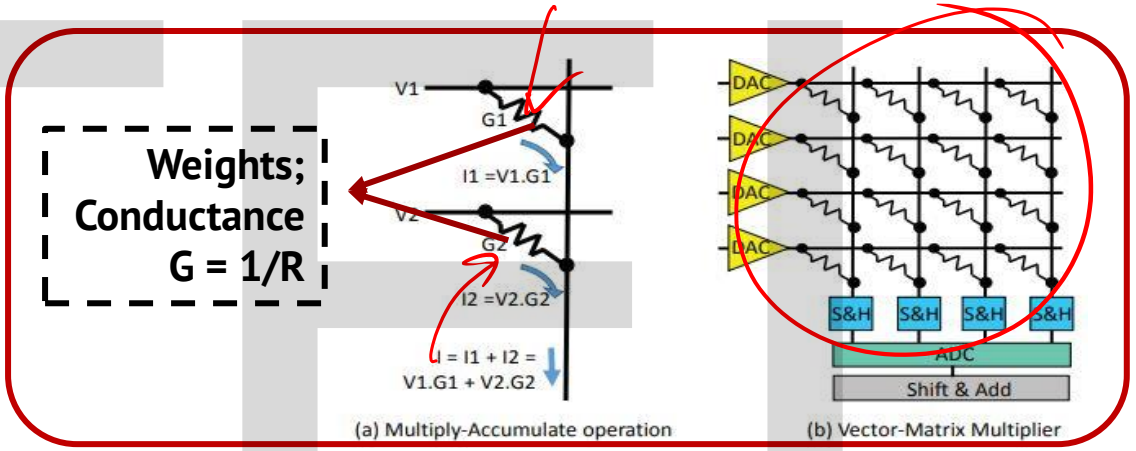
- IMC Accelerator for Transformer Algorithm
- Chiplet-based Accelerator

NPTEL

Re-Transformer: Overview

- **ReRAM-based accelerator for transformer algorithm**

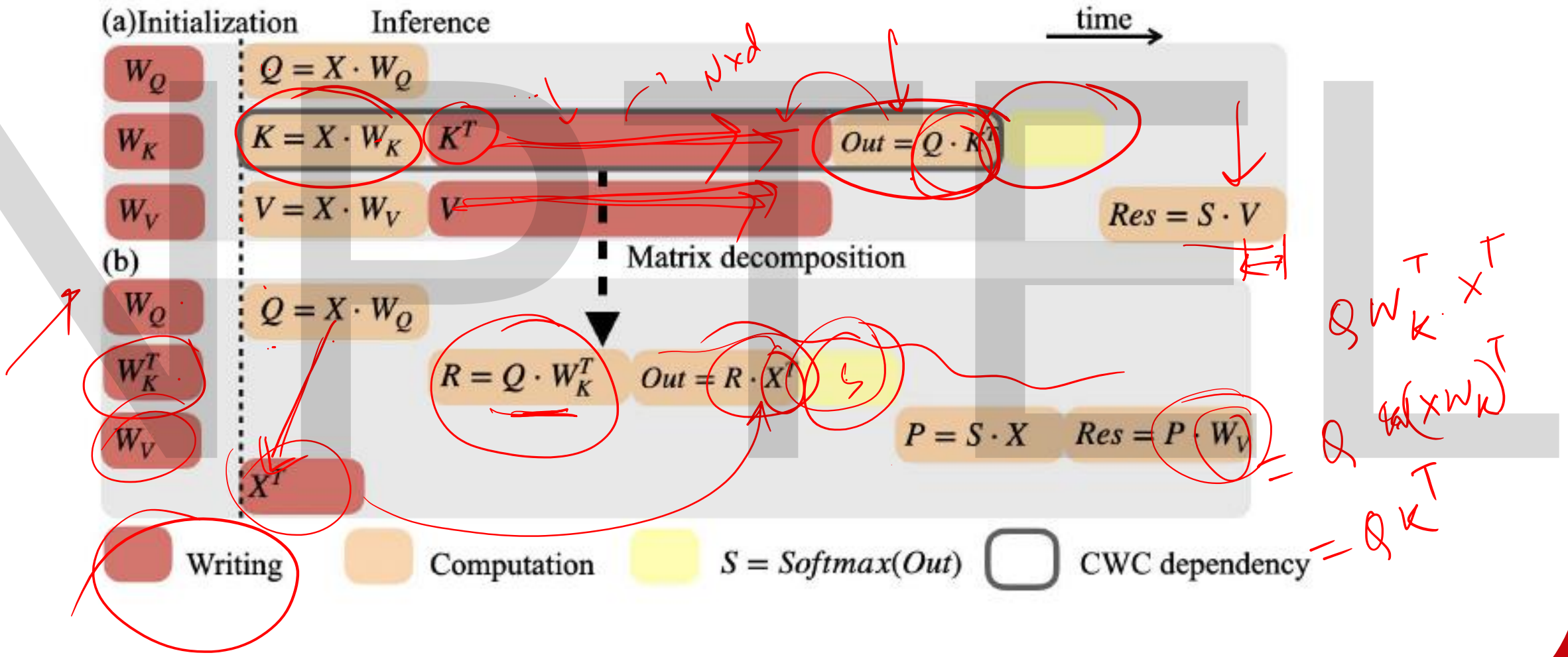




$Att = [softmax(\frac{QK^T}{\sqrt{d_K}})] \cdot V$

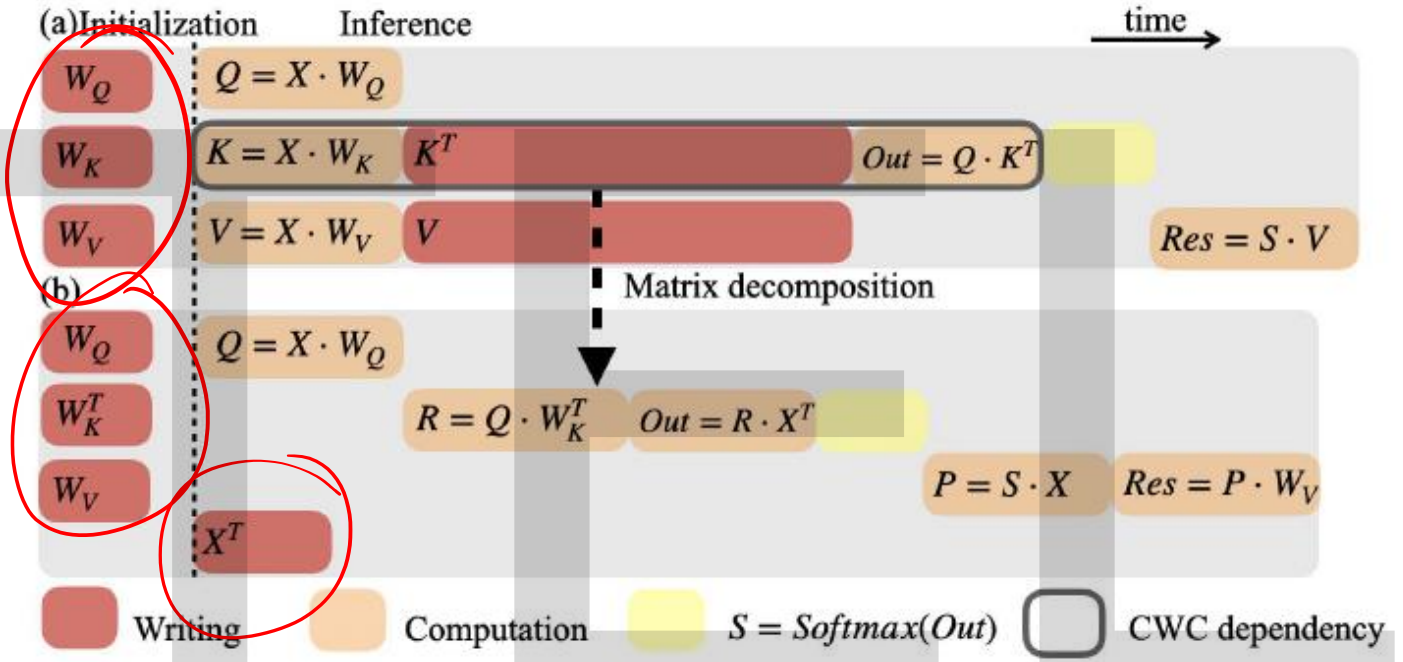
scaled dot product

ReTransformer: Reducing Memory Writes

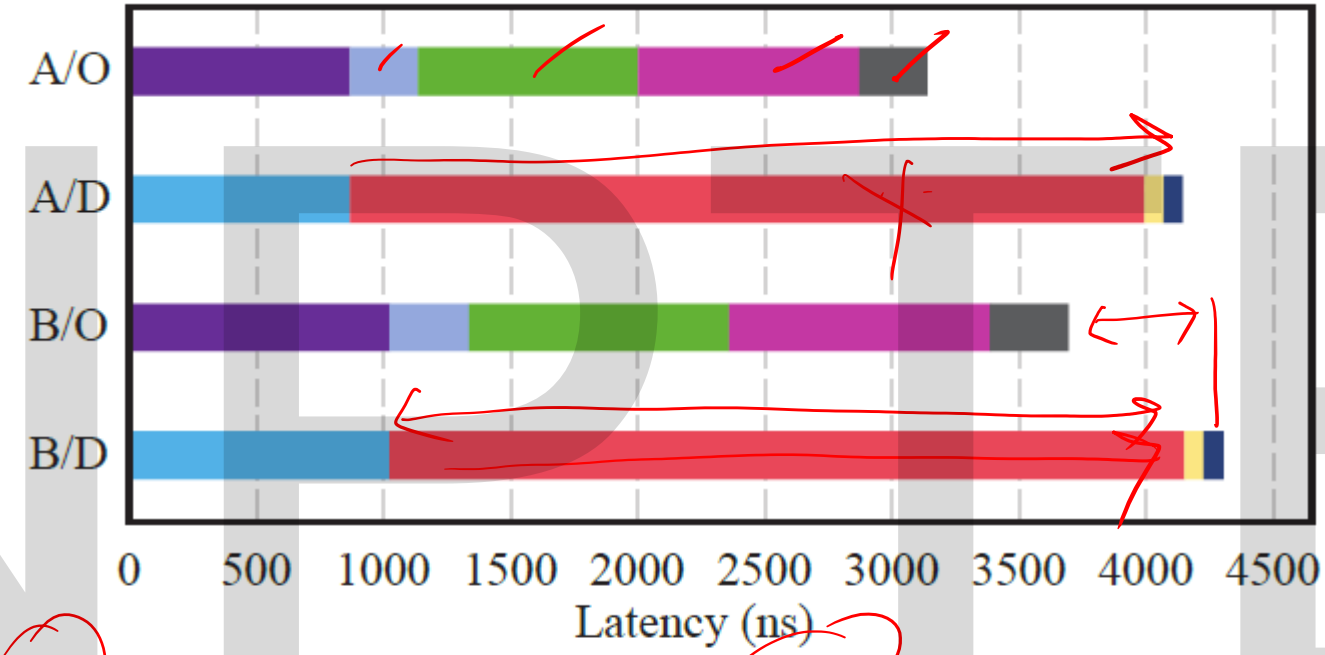


ReTransformer: Reducing Memory Writes

- Less time
- Less memory write
- Equal number of computations
- Is there any drawback?
  - More crossbar arrays



ReTransformer: Result of MatMul Optimization



A/O: Model A w/ Optimized MatMul *A/D: Model A w/ Dependency*

B/O: Model B w/ Optimized MatMul *B/D: Model D w/ Dependency*

Q and **K** Compute | **Q** Compute | $\mathbf{W_K^T}$ and **V** Write

$\mathbf{Out}=\mathbf{Q}\cdot\mathbf{W_K^T}$ Compute | **R** Compute | $\mathbf{Out}=\mathbf{R}\cdot\mathbf{X^T}$ Compute

$\mathbf{Res}=\mathbf{S}\cdot\mathbf{V}$ Compute | $\mathbf{P}=\mathbf{S}\cdot\mathbf{X}$ Compute | $\mathbf{Res}=\mathbf{P}\cdot\mathbf{W_V}$ Compute

INTEL

Agenda

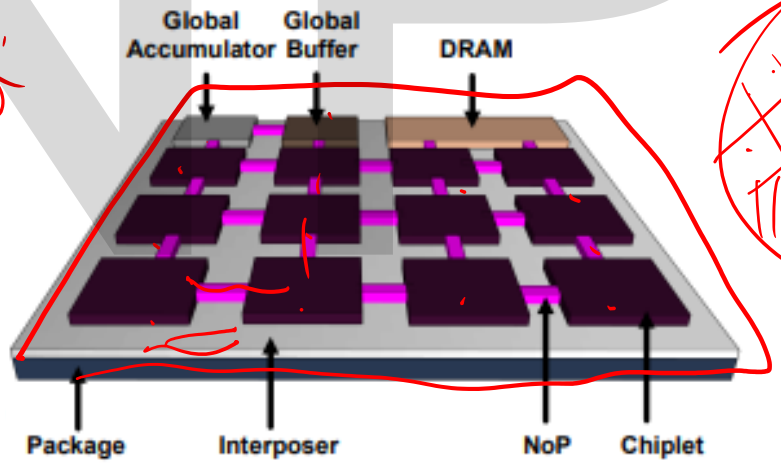
- Hardware Accelerator for Transformer Algorithm
- **Chiplet-based Accelerator [1]**

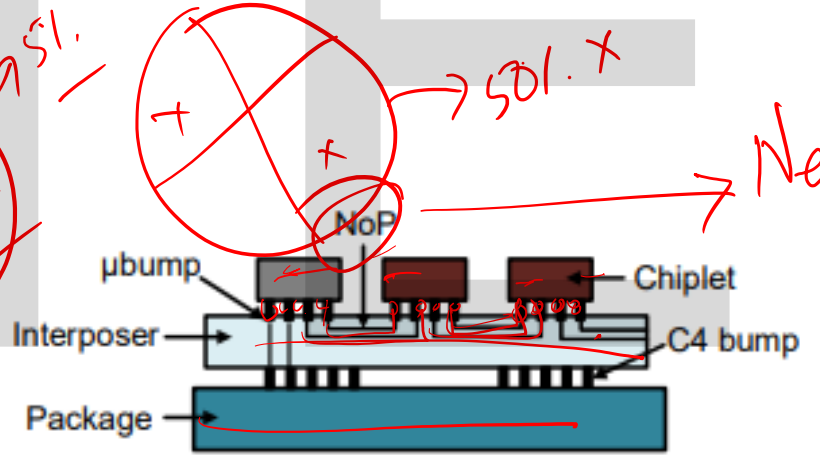
NPTEL

[1] Krishnan, Gokul, et al. "SIAM: Chiplet-based scalable in-memory acceleration with mesh for deep neural networks." ACM Transactions on Embedded Computing Systems (TECS) 20.5s (2021): 1-24.

Background (2.5D Technology)

- Yield for monolithic IC is low
- 3D IC shows thermal hotspots due to increasing number of through-silicon vias (TSV)
- 2.5D technology has emerged recently which increases yield without much of





NPTEL

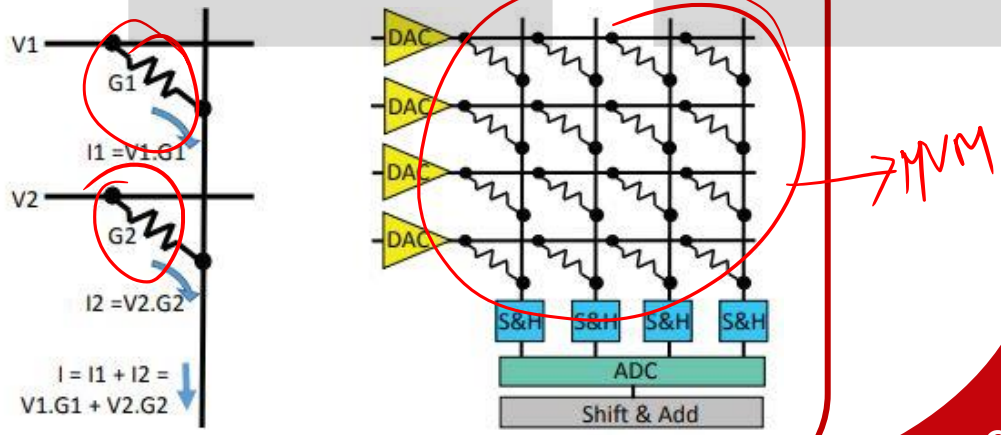
Background (In-memory Computing)

- **Analog computing: resistive (current) or capacitive (charge)**
- **Example: Resistive Random Access Memory (ReRAM or RRAM)**
  - A nonvolatile storage
  - Operates by changing the resistance of a special dielectric material
  - A family of Memristor
- **Advantage: No data movement between off chip memory**

[2] Shafiee, Ali, et al. "ISAAC: A convolutional neural network accelerator with in-situ analog arithmetic in crossbars." *ACM SIGARCH Computer Architecture News* 44.3 (2016): 14-26.

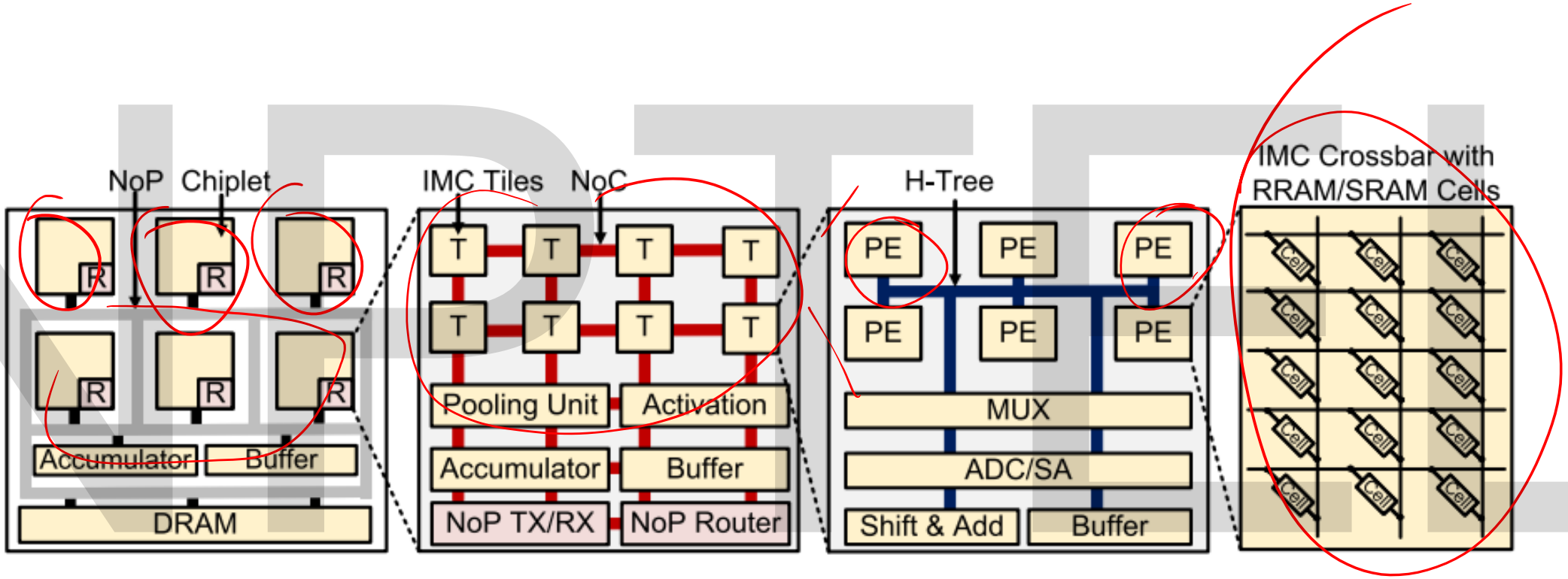
Figure adopted from [2]

Weights; Conductance G = 1/R

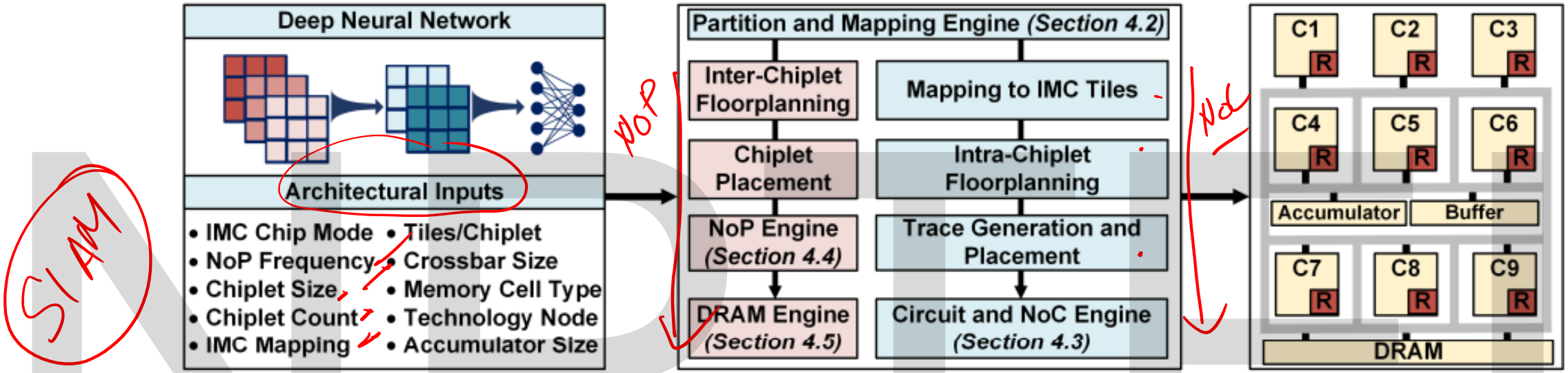


(a) Multiply-Accumulate operation (b) Vector-Matrix Multiplier

Overview of the System

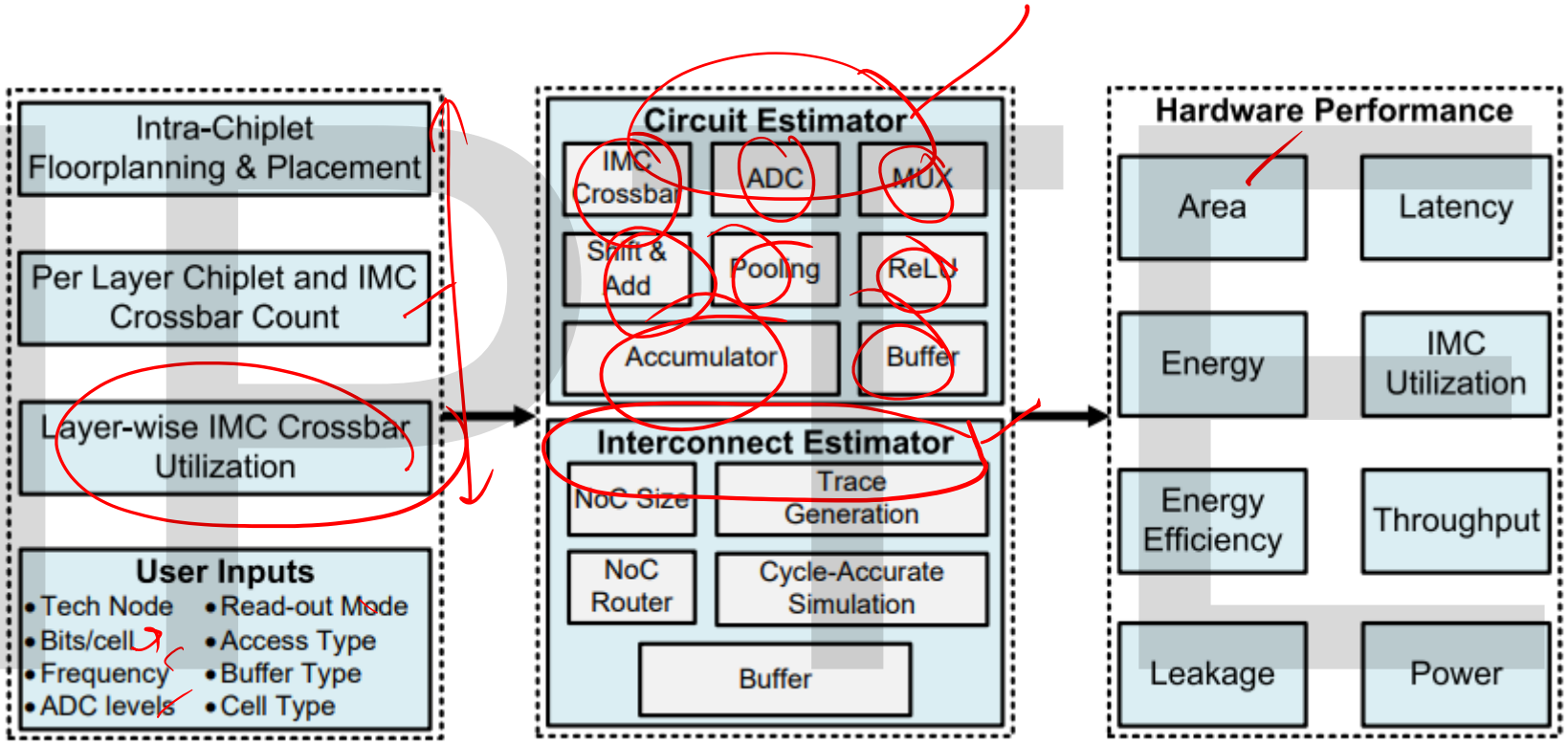


Overview of the Simulator (1)

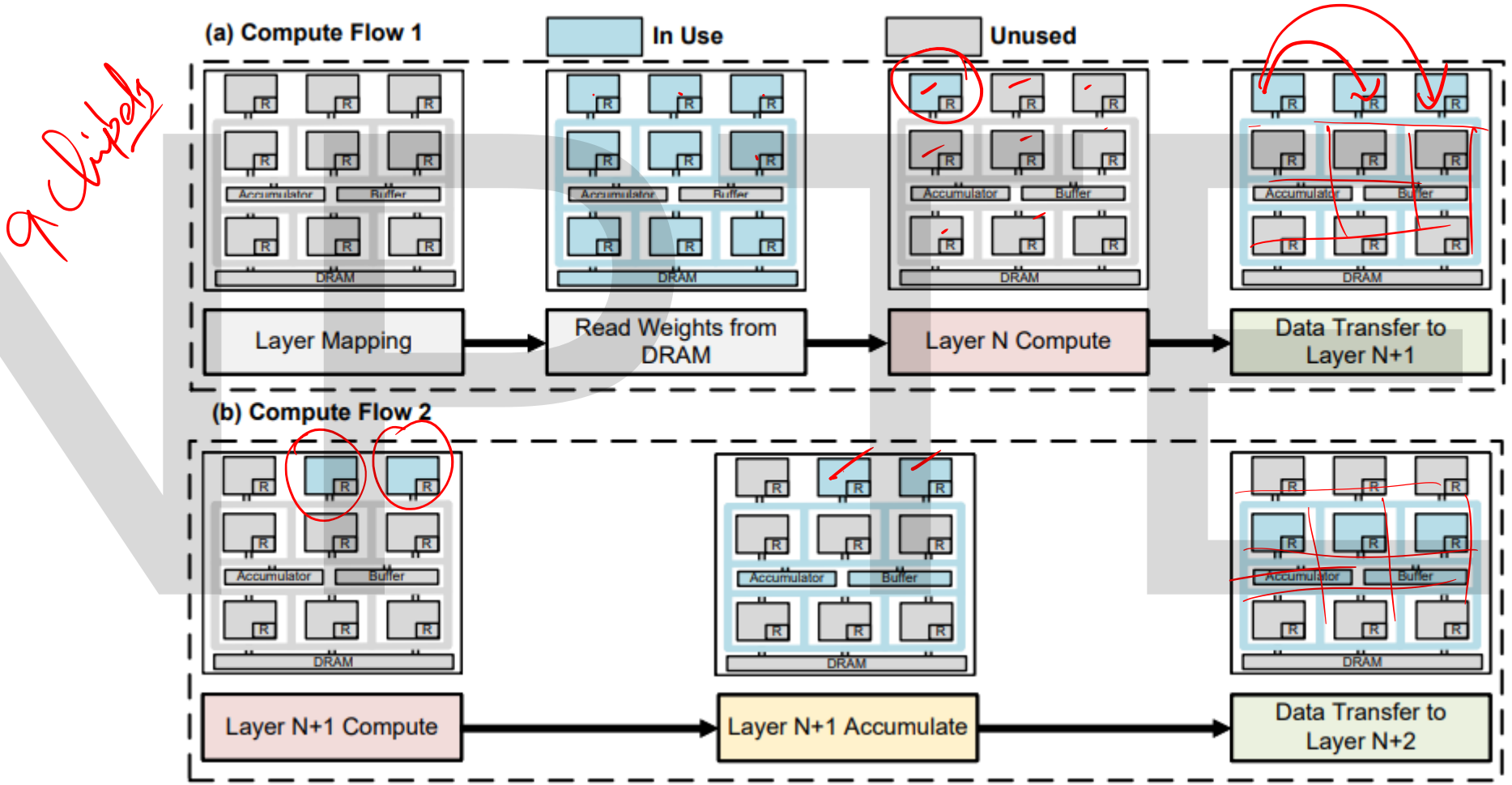


User Input	Description	User Input	Description
DNN Algorithm		**Device and Technology**	
Network Structure	DNN network structure information	Tech Node	Technology node for fabrication
Data Precision	Weights and activation precision	Memory Cell	RRAM or SRAM
Sparsity	DNN layer-wise sparsity	Bits/Cell	Number of levels in RRAM
Intra-Chiplet Architecture		**Inter-Chiplet Architecture**	
Crossbar Size	IMC crossbar array size	Chip Mode	Monolithic or chiplet-based IMC architecture
Buffer Type	SRAM or Register File	Chiplet Structure	Homogeneous or custom chiplet structure
ADC Resolution	Bit-precision of flash ADC	Chiplet Size	Number of IMC tiles within each chiplet
Read-out Method	Sequential or Parallel	Total Chiplet Count	Fixed count or DNN specific custom count
NoC Topology	Mesh or Tree	Global Accumulator Size	Size of global accumulator
NoC Width	Number of channels in the NoC	NoP Frequency	Frequency of the NoP driver and interconnect
Frequency	Frequency of operation	NoP Channel Width	Number of parallel links for TX and RX

Overview of the Simulator (2)



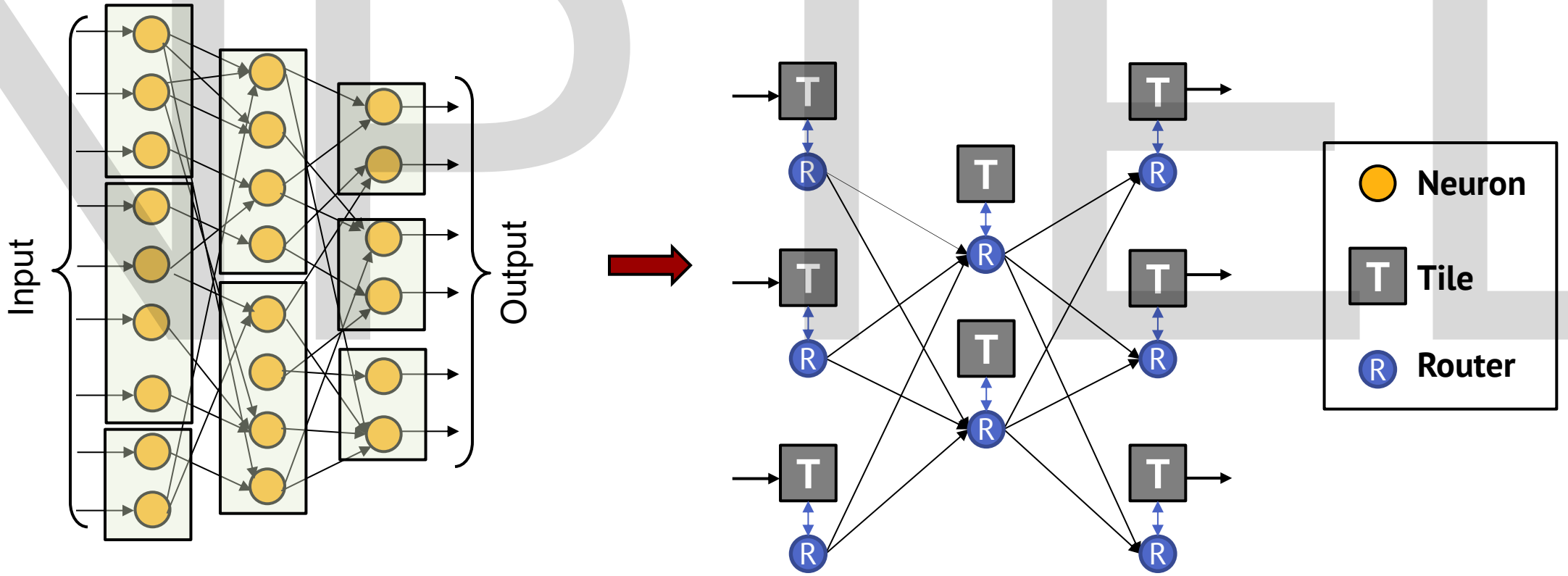
Dataflow



INTEL

Pictorial Representation

- A set of nodes are mapped to IMC tiles



NPTEL

Performance Estimation of Computation Elements

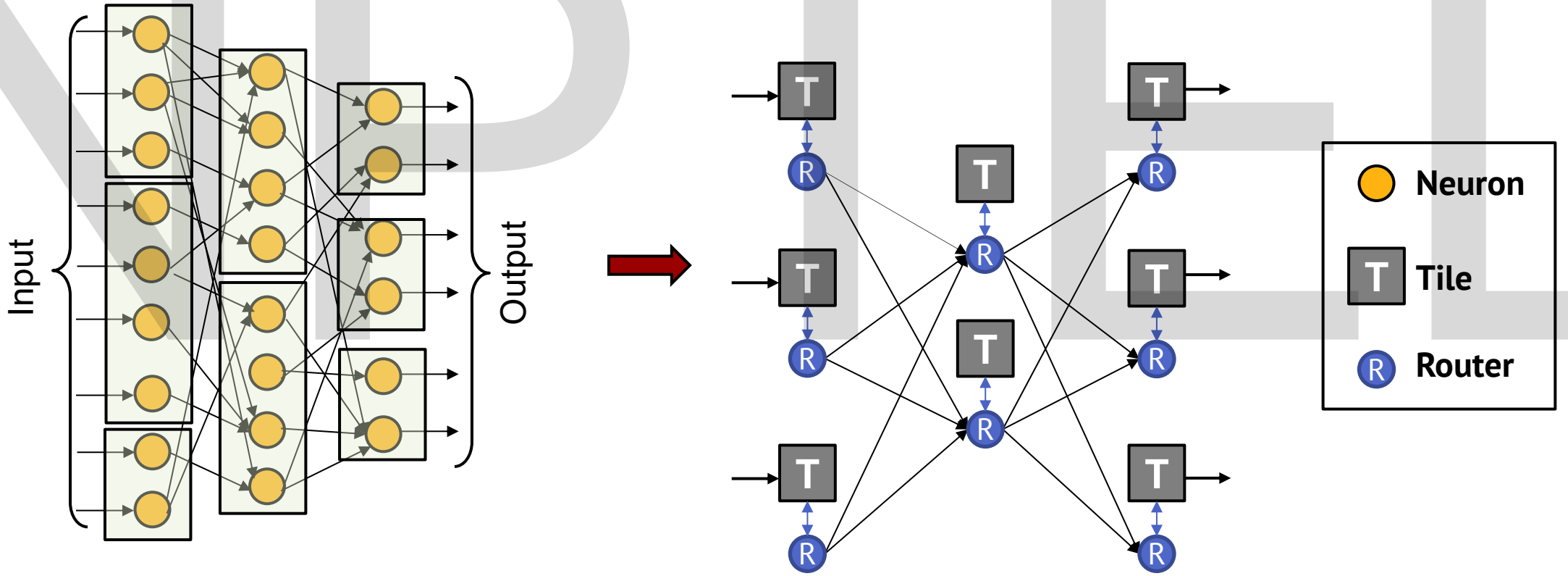
- **Modelling-based**
- **Analytical models for low-level circuit components via Predictive Technology Model (PTM) [3]**
  - Mux
  - Decoder
  - ADC/DAC
- **Required number of components are calculated**
  - Performance is estimated via scaling
- **Fast; since equations are executed again and again**

NPTEL

[3] Zhao, Wei, and Yu Cao. "Predictive technology model for nano-CMOS design exploration." ACM Journal on Emerging Technologies in Computing Systems (JETC) 3.1 (2007): 1-es.

Pictorial Representation (Revisited)

- A set of nodes are mapped to IMC tiles



NPTEL

Performance Estimation of Communication

- **Typically, a block of computation is performed at once**
- **Many packets are ready to be communicated at a single cycle**
  - No particular distribution in arrival rate
- **Need to know the total latency**
  - Time taken by the communication to be finished
  - Average latency is not important
- **Deploying queuing theory-based analytical model is not feasible**
- **Cycle accurate simulation is performed to estimate communication performance**

NPTEL

Step 1: Trace File Generation

- **Trace file contains below information**
  - **Source**
  - **Destination**
  - **Time stamp**

Algorithm 2: NoC (or NoP) Trace Generation

Input: Number of tiles for each layer (for each chiplet in case of NoP) ($|\mathcal{T}|$), Number of input activations for each layer(A), Number of chiplets (C), Layer to chiplet mapping ($\mathcal{L} \rightarrow C$), Quantization bit ($Q$), Bus width ($W$)
Output: Trace file for each chiplet (tr^c)
for $c = 1 : |C|$ **do**
  Find index of the first layer (L_c^S) and the last layer (L_c^E) in the chiplet from $\mathcal{L} \rightarrow C$
  for $l = L_c^S : L_c^E$ **do**
    $k = 0$ /* **Initialize timestamp** */
    Find index of first source tile (T_l^S) and last source tile (T_l^E)
    Find index of first destination tile (T_{l+}^S) and last destination tile (T_{l+}^E)
    $N_p = \lceil \frac{A(l)Q}{W} \rceil$ /* **Number of packets** */
    for $n = 1 : N_p$ **do**
      for $s = T_l^S : T_l^E$ **do**
        for $d = T_{l+}^S : T_{l+}^E$ **do**
          $tr^c \leftarrow [tr^c; [s, d, k]]$
          $k \leftarrow k + 1$ /* **Increment timestamp** */
        end
        $k \leftarrow k + 1$ /* **Increment timestamp** */

NPTEL

Example of a Trace File

- Three columns
  - Source router
  - Destination router
  - Time stamp
- Routers cannot send more than one packet at once cycle
- Number of lines in the trace file represents number of packets

```
0 1 1
2 1 1
4 1 1
0 1 2
2 1 2
4 1 2
0 1 3
2 1 3
4 1 3
```

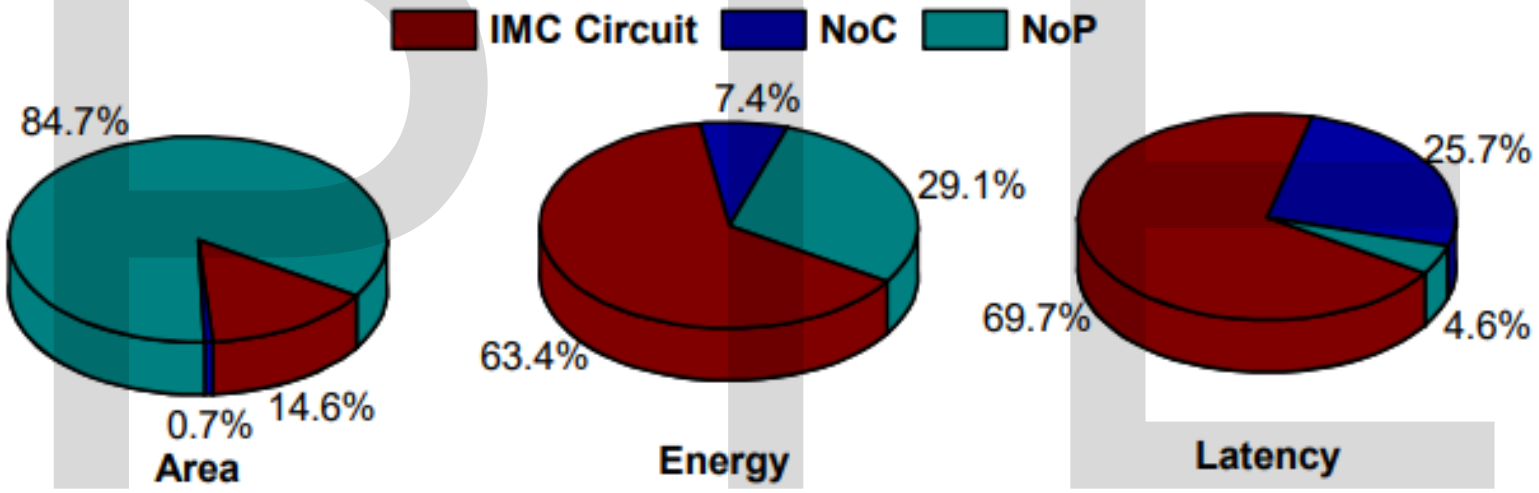
NPTEL

Simulating the Trace File

- Cycle accurate simulation
  - BookSim
- Mesh interconnect
- The timestamp when all the packets are received is collected
- BookSim also provides area and energy estimation

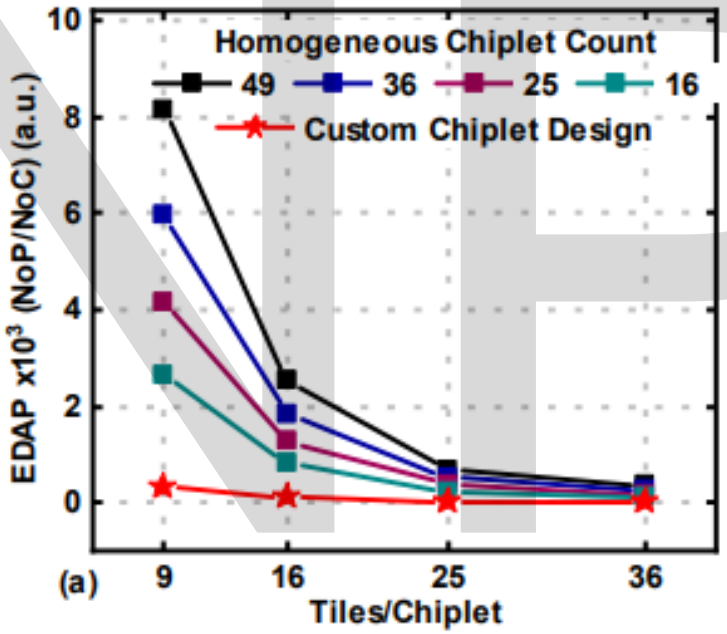
NPTEL

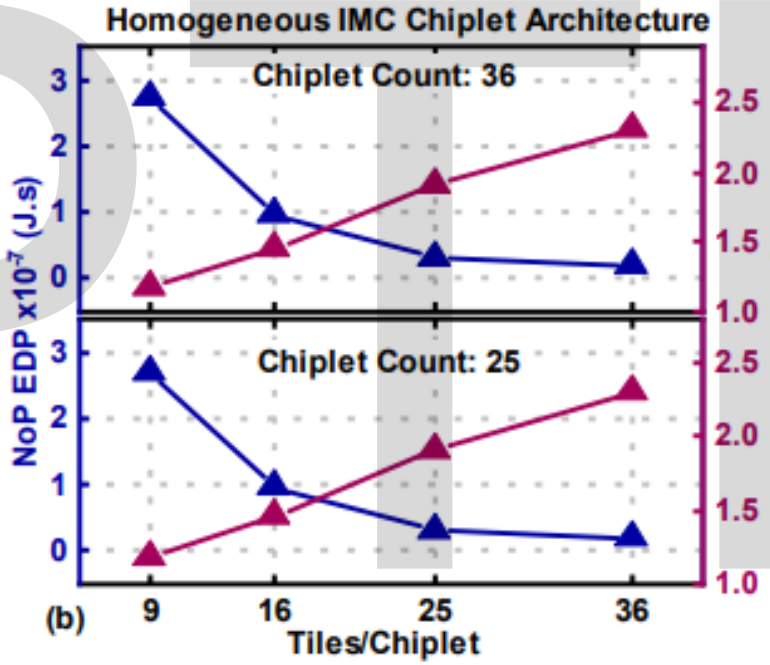
Results (1)

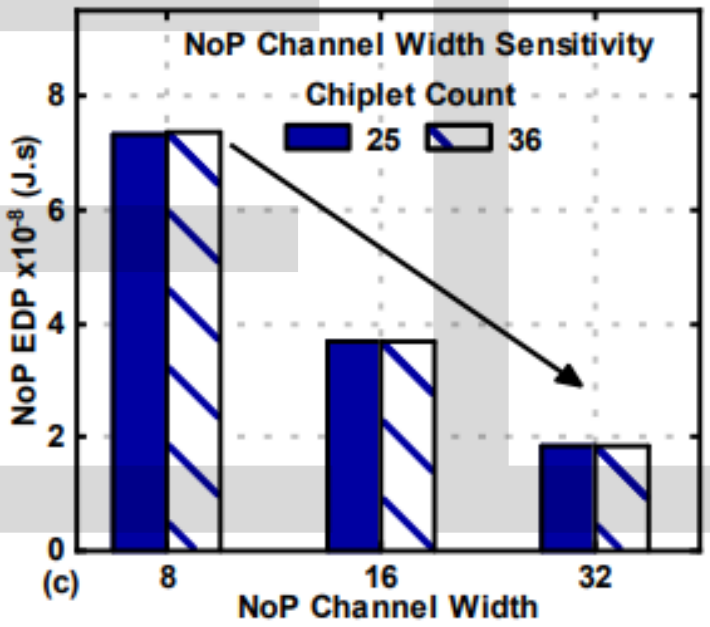


NPTEL

Results (2)

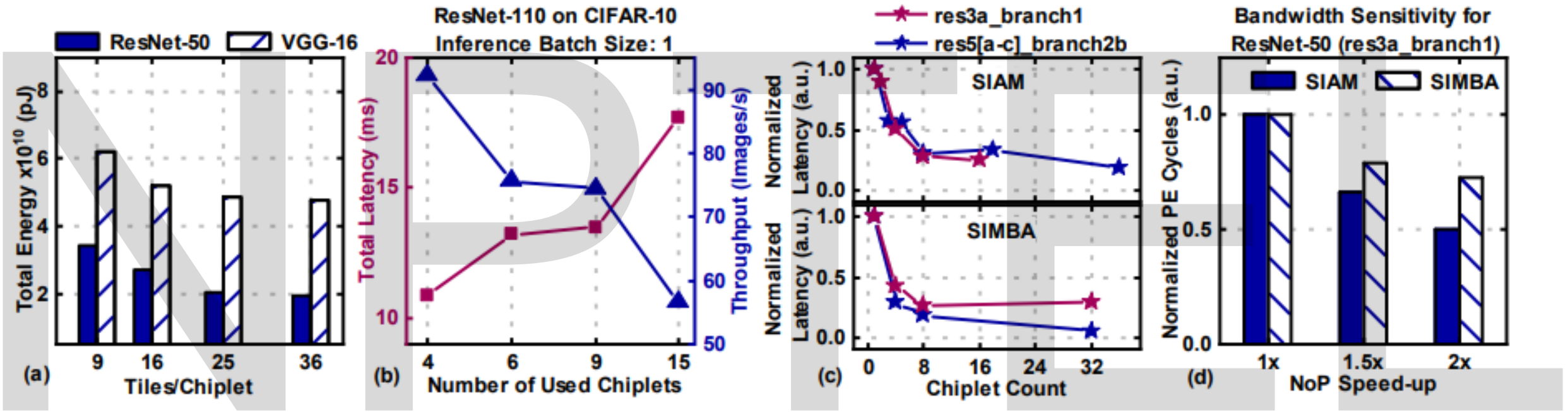






NDA

Results (3)



More about hardware accelerator for machine learning applications will be discussed in the 'E0294: Systems for Machine Learning' course in the next semester



THANK YOU

Slide Courtesy: Murali Annavaram (University of Southern California), R. Govindarajan (IISc), Arkaprava Basu (IISc), Jayashree Mohan (Microsoft Research)